The documentation and process conversion measures necessary to comply with this revision shall be completed by 21 March 1994

INCH-POUND

MIL-S-19500/115H
20 December 1993
SUPERSEDING
MIL-S-19500/115G
2 April 1990

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, VOLTAGE REGULATOR TYPES
1N3821A THROUGH 1N3828A, 1N3016B THROUGH 1N3051B,
1N3821A-1 THROUGH 1N3828A-1, 1N3016B-1 THROUGH 1N3051B-1,
1N3821AUR-1 THROUGH 1N3828AUR-1, 1N3016BUR-1 THROUGH 1N3051BUR-1,
PLUS C- AND D- TOLERANCE SUFFIX, JAN, JANTX, JANTXV, AND JANHC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the detail requirements for 1 Watt, silicon, voltage regulator diodes with voltage tolerances of 5 percent (2 percent and 1 percent tolerance options are available on the -1 suffix devices and JANHC). Three levels of product assurance are provided for each device type as specified in MIL-S-19500. One level of product assurance is provided for die.

1.2 Physical dimensions. See figures 1 (DO-13), 2 (similar to DO-41), 3 and 4 (JANHC).

1.3 Maximum ratings. Maximum ratings are as shown in columns 8 and 10 of table IV herein and as follows:

$P_T$ = 1.0 W (DO-13 and DO-41) at $T_L$ = 95°C, L = .375 (9.53 mm); both ends of case or diode body to heat sink at L = .375 (9.53 mm). (Derate $I_Z$ to 0.0 mA dc at $T_L$ = +175°C).

$P_T$ = 1.0 W (DO-213AB) at $T_{EC}$ = 125°C. (Derate to 0 at $T_{EC}$ = 175°C).

-65°C ≤ $T_{op}$ ≤ +175°C; -65°C ≤ $T_{STG}$ ≤ +175°C.

1.4 Primary electrical characteristics. Primary electrical characteristic columns 2, 9, 12, and 15 of table IV herein and as follows:

3.3 V dc ≤ $V_Z$ ≤ 200 V dc
1N3821D-1 through 1N3828D-1 and 1N3016D-1 through 1N3051D-1 are 1 percent voltage tolerance.
1N3821C-1 through 1N3828C-1 and 1N3016C-1 through 1N3051C-1 are 2 percent voltage tolerance.
1N3821A through 1N3828A and 1N3016B through 1N3051B are 5 percent voltage tolerance.
1N3821A-1 through 1N3828A-1 and 1N3016B-1 through 1N3051B-1 are 5 percent voltage tolerance.

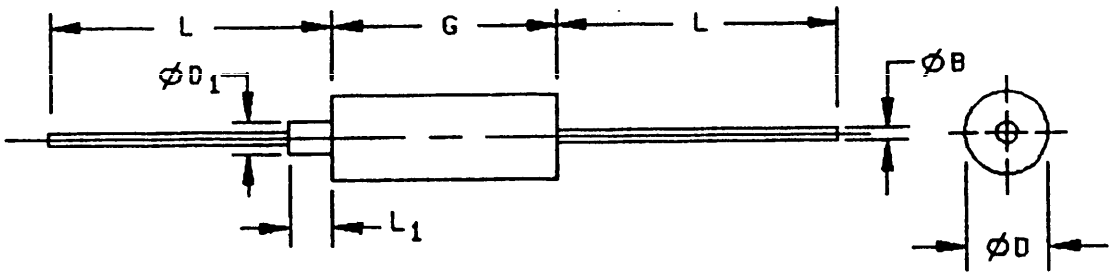
$R_{\Theta JL}$ = 80°C/W (maximum) at L = .375 inch (9.53 mm) (DO-13 and DO-41).

$R_{\Theta JEC}$ = 50°C/W (maximum) junction to endcaps (DO-213AB).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Electronics Supply Center, ATTN: DESC-ECT, 1507 Wilmington Pike, Dayton, OH 45444-5270, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.


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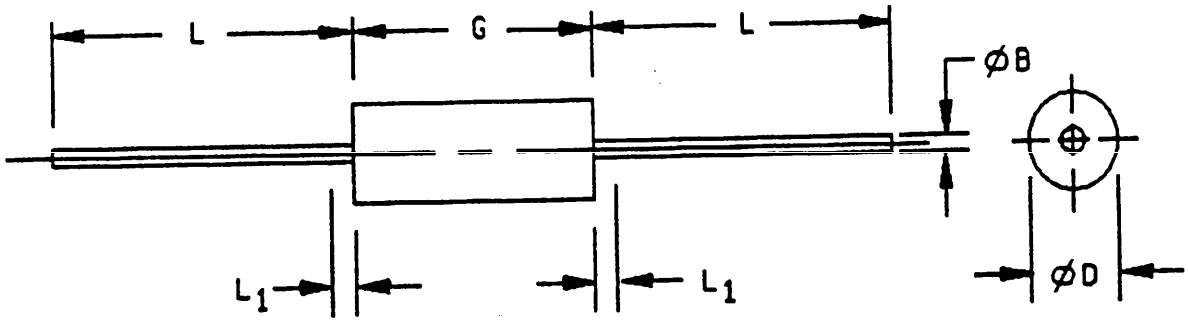

| Ltr | Dimensions | | | | Notes |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| ϕB | .026 | .035 | 0.66 | 0.89 | |
| ϕD | .215 | .265 | 5.46 | 6.73 | 4 |
| ϕD1 | | .110 | | 2.79 | |
| G | .195 | .350 | 4.96 | 8.89 | |
| L | 1.000 | | 25.40 | | |
| L1 | | .21 | | 5.33 | |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimension ϕD shall be measured at the largest diameter.
4. Cathode lead shall be electrically connected to the case. If tubulation is used, it shall be on the anode end.

FIGURE 1. Semiconductor device, diode, types 1N3821A through 1N3828A, and 1N3016B through 1N3051B.



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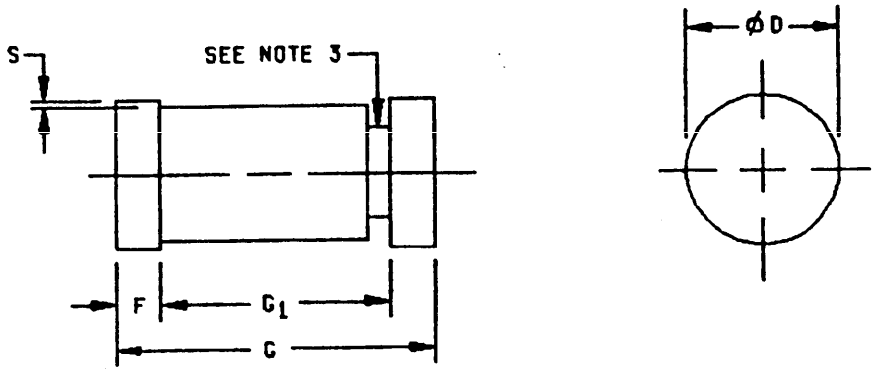

| Ltr | Dimensions | | | | |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| ϕB | .028 | .034 | 0.71 | 0.86 | |
| ϕD | .080 | .107 | 2.03 | 2.72 | 3 |
| G | .160 | .205 | 4.06 | 5.21 | 3 |
| L | 1.110 | --- | 28.19 | --- | |
| L1 | --- | .050 | --- | 1.27 | 4 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Package contour optional within cylinder of diameter ϕD and length G. Slugs, if any shall not be included within this cylinder but shall not be subject to the minimum limit of ϕD.
4. Lead diameter not controlled in this zone to allow for flash, lead finish build-up, and minor irregularities other than slugs.

FIGURE 2. Physical dimensions, types 1N3821A-1, C-1, and D-1 through 1N3828A-1, C-1, and D-1; 1N3016B-1, C-1, and D-1 through 1N3051B-1, C-1, and D-1 (DO-41).



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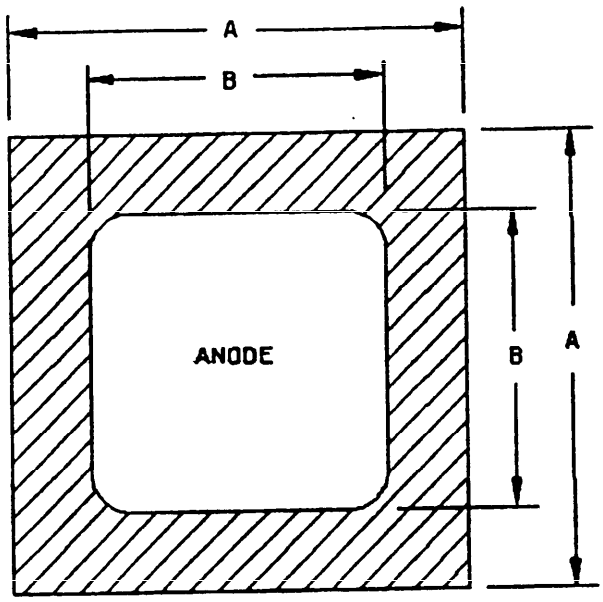

| Ltr | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| G | .189 | .205 | 4.80 | 5.21 |
| ϕD | .094 | .105 | 2.39 | 2.67 |
| G1 | .159 REF. | | 4.04 REF. | |
| F | .014 | .022 | .36 | .56 |
| S | .001 | --- | .03 | --- |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Gap not controlled, shape of body and gap not controlled.

FIGURE 3. Physical dimensions of surface mount family types 1N3821AUR-1, CUR-1, and DUR-1, through 1N3828AUR-1, CUR-1, and DUR-1 and 1N3016BUR-1, CUR-1, and DUR-1 through 1N3051BUR-1, CUR-1, and DUR-1 (DO-213AB).



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BACKSIDE IS CATHODE

**A Version**

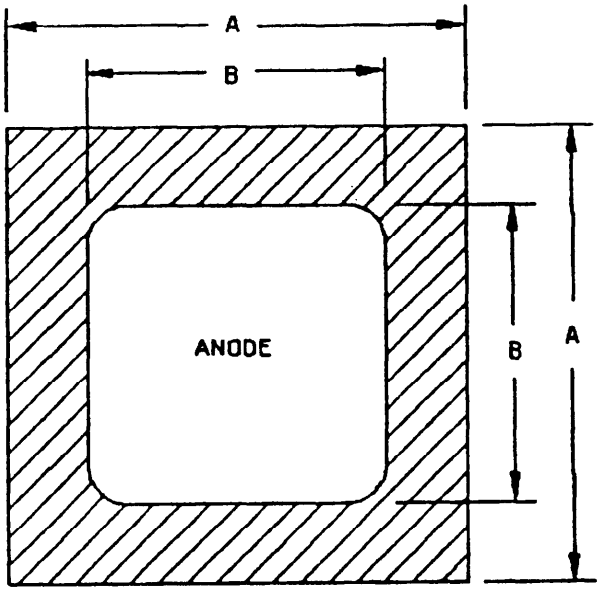
| Ltr | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| A | .035 | .039 | .89 | .99 |
| B | .031 | .033 | .79 | .84 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The physical characteristics of the die thickness are .010 .±002 (0.25 mm). Metallization is: Top (anode) - Al, back (cathode) - Au. Al thickness = 25,000Å minimum, Au thickness = 4000Å minimum.

FIGURE 4. Physical dimensions JANHCA die.



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BACKSIDE IS CATHODE

B Version

| Ltr | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| A | .035 | .039 | 0.89 | 0.99 |
| B | .027 | .031 | 0.68 | 0.79 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The physical characteristics of the die thickness are .012 .±002 (0.30 mm). Metallization is: Top (anode) - Al, back (cathode) - Au. Al thickness = 40,000Å minimum, Au thickness = 5000Å minimum.

FIGURE 5. Physical dimensions JANHCB die.



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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions shall be as specified in MIL-S-19500.

EC - - - - - - - - - - - - - - - - - - - - endcap.

3.3 Design, construction, and physical dimensions. Design, construction, and physical dimensions shall be as specified in MIL-S-19500, and figures 1 (DO-13), 2 (DO-41), 3 (DO-213AB), and figures 4 and 5 for (JANHC).

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-STD-750 and MIL-S-19500 where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.3.2 Dash one construction. Dash one (-1) diodes shall be of metallurgically bonded double plug construction in accordance with MIL-S-19500.

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

3.4.1 Marking of UR version devices. For UR version devices only, all marking (except polarity) may be omitted from the body, but shall be retained on the initial container.

3.4.2 Polarity. The polarity shall be indicated with a contrasting color band to denote the cathode end or alternately with a minimum of three contrasting color dots spaced evenly around the periphery at the cathode end.



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3.5 Selection of tight tolerance devices. The C and D suffix devices shall be selected from JAN, JANTX, or JANTXV devices, which have successfully completed all applicable screening, and groups A, B, and C testing as 5 percent tolerance devices. All sublots of C and D suffix devices shall pass group A, subgroup 2, at tighter tolerances. Tighter tolerances for mounting clip temperature shall be maintained for reference purpose to establish correlation. For C and D tolerance levels, $T_L$ = 30 ± 2°C at 0.375" from body or equivalent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2.1 JANHC devices. JANHC devices shall be qualified in accordance with appendix H of MIL-S-19500.

4.2.2 Construction verification. Cross sectional photos from 3 devices shall be submitted in the qualification report.

4.3 Screening (JAN, JANTXV and JANTX levels only). Screening shall be in accordance with MIL-S-19500 (table II), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see table II of MIL-S-19500) | Measurement | |
|---|---|---|
| | JANTX and JANTXV levels | JAN Level |
| 3a | Temperature cycling | Temperature cycling in accordance with MIL-S-19500 JANTX level) |
| 3c 1/ | Thermal impedance (see 4.5.5) | Thermal impedance (see 4.5.5) |
| 9 | Not applicable | Not applicable |
| 11 | $I_R$ and $V_Z$ | |
| 12 | See 4.3.2,t = 48 hours | |
| 13 2/ | $\Delta I_{R1}$ ≤ 100% of initial reading or 50 nA dc, whichever is greater; $\Delta V_Z$ ≤ ±2% of initial reading, subgroup 2 of table I herein. | |

1/ Thermal impedance may be performed any time after sealing provided temperature cycling is performed in accordance with MIL-S-19500, screen 3 prior to this thermal test.
2/ PDA = 5 percent for screen 13, applies to $\Delta I_{R1}$, $\Delta V_Z$ and subgroup 2 of table I herein. Thermal impedance ($Z_{\theta JX}$) is not required in screen 13.

4.3.1 Screening (JANHC). Screening of JANHC die shall be in accordance with MIL-S-19500, appendix H. As a minimum, die shall be 100-percent probed in accordance with group A, subgroup 2.



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4.3.2 Power burn-in conditions. Power burn-in conditions are as follows: $I_Z$ = column 8 of table IV minimum; mounting and test conditions in accordance with MIL-STD-750, method 1038, test condition B, TEC = +75°C to +125°C for surface mount devices. To better utilize burn-in equipment, higher values of $I_Z$ shall be permitted provided:

a. The junction temperature does not exceed +175°C.
b. The power dissipation does not exceed 1.2 W.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein. The following test conditions shall be used for $Z_{\theta JX}$, group A inspection:

a. $I_M$ measurement current - - - - - - - - - - - - - - - - 1 mA - 10 mA.
b. $I_H$ forward heating current - - - - - - - - - - - - - - 1.0 A - 2.0 A.
c. $t_H$ heating time - - - - - - - - - - - - - - - - - - - - 10 ms.
d. $t_{MD}$ measurement delay time - - - - - - - - - - - - - 70 µs maximum.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IIb (JAN, JANTX and JANTXV) of MIL-S-19500, and as follows. Electrical measurements (endpoints) and delta requirements shall be in accordance with the applicable footnotes and steps of table III herein.

| Subgroup | Method | Condition |
|---|---|---|
| B2 | 4066 | See 4.5.1 |
| B3 | 1027 | $I_Z = I_{ZM}$ column 8 of table IV; $T_A$ = +30° ±5°C. |
| B4 | ---- | Not applicable |
| B5 | ---- | Not applicable |

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and as follows. Electrical measurements (endpoints) and delta requirements shall be in accordance with the applicable footnotes and steps of table III herein.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | Terminal strength: Test condition A; weight = 4 lbs; t = 15 seconds.<br>Terminal strength: Test condition E;<br>(Terminal strength is not required for UR-1 suffix devices). |
| C6 | 1026 | $I_Z = I_Z$ column 8 of table IV; $T_A$ = +30°C ±5°C. |
| C7 | 4071 | $I_Z = I_Z$ column 5 of table IV; $T_A$ = +25°C ±5°C; $T_2$ = +125°C ± 5°C; $\alpha V_Z$ = column 15 of table IV; LTPD = 10. |

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Surge current ($I_{ZSM}$). The peak currents shown in column 10 of table IV shall be applied in the reverse direction and these shall be superimposed on the current ($I_Z = I_{Z1}$)(column 5 of table IV) a total of 5 surges at 1 minute intervals. Each individual surge shall be one-half square-wave-pulse of one one-hundred twentieth second duration or an equivalent one-half sinewave with the same effective rms current.



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4.5.2 Regulator voltage measurements. The test current shall be applied until thermal equilibrium is attained (90 seconds maximum) prior to reading the breakdown voltage. For this test, the surface mount device shall be mounted at the endcaps and the axial leaded device shall be suspended by its leads with mounting clips whose inside edge is located at 0.375 inch (9.53 mm) from the body and the mounting clips shall be maintained at a temperature of +25°C +8°C, -2°C . This measurement may be performed after a shorter time following application of the test current than that which provides thermal equilibrium if correlation to stabilized readings can be established to the satisfaction of the Government.

4.5.3 Temperature coefficient of regulator voltage ($\alpha V_Z$). The device shall be temperature stabilized with current applied prior to reading regulator voltage at the specified ambient temperature as specified in table III, group C, subgroup 7.

4.5.4 Voltage regulation $V_Z$ (reg). Voltage regulation shall be determined by the difference of the regulator voltage measured at different currents as specified in table I, group A, subgroup 7. Both tests shall be performed at thermal equilibrium. This $\Delta V_Z$ shall not exceed column 9 of table IV.

4.5.5 Thermal impedance $Z_{\theta JX}$ measurements for screening. The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101. The maximum limit (not to exceed the group A, subgroup 2 limit) for $Z_{\theta JX}$ in screening (table II of MIL-S-19500) shall be derived by each vendor by means of statistical process control. When the process has exhibited control and capability, the capability data shall be used to establish the fixed screening limit. In addition to screening, once a fixed limit has been established, monitor all future sealing lots using a random five piece sample from each lot to be plotted on the applicable X, R chart. If a lot exhibits an out of control condition, the entire lot shall be removed from the line and held for Engineering evaluation and disposition.

4.5.5.1 For initial qualification or requalification. Read and record data ($Z_{\theta JX}$)shall be supplied to the qualifying activity on one lot (random sample of 500 devices minimum). Twenty-two serialized devices shall be sent to the qualifying activity for test correlation.

4.5.6 Thermal resistance. Thermal resistance measurement shall be in accordance with MIL-STD-750, method 3101 or 4081. Forced moving air or draft shall not be permitted across the device during test. The maximum limit for $R_{\theta JL}$ under these test conditions shall be $R_{\theta JL}$ (max) = 80°C/W or $R_{\theta JEC}$ = 50°C/W. The following conditions shall apply when using method 3101:

a. $I_M$ - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - 1 mA to 10 mA
b. $I_H$ - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - .5 A to 1.0 A
c. $t_H$ - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - 25 seconds minimum
d. $t_{MD}$ - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - 70 µs maximum

LS = Lead spacing = 3/8 inch as defined on figure 6 below:
LS = 0 inches for "UR" suffix devices.

4.5.6.1 For initial qualifications and requalifications. Read and record data in accordance with group E herein and shall be included in the qualification report.

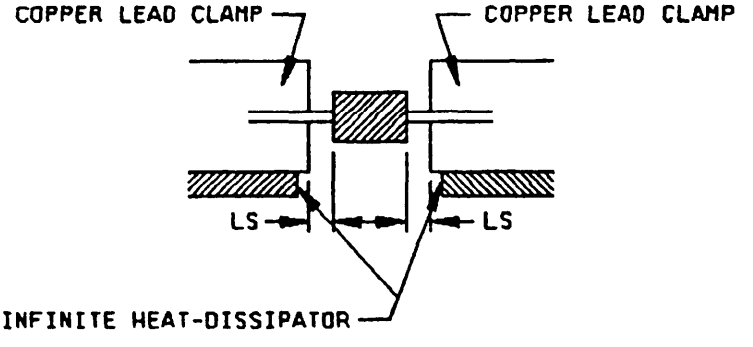



FIGURE 6. Mounting conditions.



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TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | 2/ Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical examination | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| Forward voltage | 4011 | $I_F$ = 200 mA dc | $V_F$ | | 1.2 | V dc |
| Reverse current | 4016 | DC method; $V_R$ = column 11 of table IV | $I_{R1}$ | | Column 12 | µA dc |
| Regulator voltage (see 4.5.2) | 4022 | $I_{Z1}$ = $I_Z$ = (column 5 of table IV) | $V_Z$ | Column 3 | Column 4 | V dc |
| Thermal impedance | 3101 | See 4.5.5 | $Z_{\theta JX}$ | | 15 | °C/W |
| Subgroup 3 | | (-1 devices only) | | | | |
| High-temperature operation | | $T_A$ = +150°C | | | | |
| Reverse current | 4016 | DC method: $V_R$ = column 11 of table IV | $I_{R2}$ | | Column 14 | µA dc |
| Subgroup 4 | | | | | | |
| Small-signal reverse breakdown impedance | 4051 | $I_Z$ = column 5 of table IV. $I_{sig}$ = 10 % of $I_Z$ | $Z_Z$ | | Column 6 | ohms |
| Small-signal knee impedance | 4051 | $I_{ZK}$ = column 16 of table IV; $I_{sig}$ = 10 % of $I_{ZK}$ | $Z_{ZK}$ | | Column 7 | ohms |
| Subgroups 5, and 6 | | | | | | |
| Not applicable | | | | | | |
| Subgroup 7 | | | | | | |
| Voltage regulation (see 4.5.4) | | $I_Z$ = 10% of column 8 of table IV (current 1) $I_Z$ = 50% of column 8 of table IV (current 2) | $\Delta V_Z$ (reg) | | Column 9 | V dc |

1/ For sampling plan, see MIL-S-19500
2/ Column references are to table IV herein.



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TABLE II. Group E inspection (all product assurance levels).

| Inspection 1/ | MIL-STD-750 | | Qualification conformance inspection (LTPD) |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | |
| Temperature cycling | 1051 | 500 cycles | 45/0 |
| Electrical measurements | | See table III, steps 1, 2, 3, and 5. | |
| Subgroup 2 | | | |
| Steady-state dc intermittent life | 1037 | 6,000 cycles | 45/0 |
| Electrical measurements | | See table III, steps 1, 2, and 3. | |
| Subgroup 3 | | | |
| Not applicable | | | |
| Subgroup 4 | | | 22 devices, c = 0 |
| Thermal resistance surface mount | 3101 or 4081 | $R_{\theta JEC}$ = 50°C/W (maximum) at zero lead length. +25°C ≤ $T_R$ ≤ +35°C, (see 4.5.6). | |
| Thermal resistance leaded | 3101 or 4081 | $R_{\theta JL}$ = 80°C/W (maximum) at .375 inches lead length +25°C ≤ $T_R$ ≤ 35°C, (see 4.5.6). | 22 devices, c = 0 |

1/ A separate sample may be pulled for each test.



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TABLE III. Group A, B, C, and E electrical end-point measurements. 1/, 2/, 3/

| Step | Inspection | MIL-STD-750 | | Symbol | Limits 1/ | | Unit |
|---|---|---|---|---|---|---|---|
| | | Method | Conditions | | Min | Max | |
| 1. | Reverse current | 4016 | DC method, $V_R$ = column 11 of table IV | $I_{R1}$ | | Column 13 | µA dc |
| 2. | Breakdown voltage (see 4.5.2) | 4022 | $I_Z$ = column 5 of table IV | $V_Z$ | Column 3 | Column 4 | V dc |
| 3. | Small-signal breakdown impedance | 4051 | $I_Z$ = column 5<br>$I_{sig}$ = 10% of $I_Z$ | $Z_Z$ | | Column 6 | ohms |
| 4. | Thermal impedance | 3101 | See 4.5.5 | $Z_{\theta JX}$ | | 15 | °C/W |

1/ Column references are to table IV herein.

2/ The electrical measurements for table IVb (JAN, JANTX, and JANTXV) of MIL-S-19500 are as follows:

a. Subgroup 2, see table III herein, steps 1, 2, 3, and 4.

b. Subgroups 3 and 6, see table III herein, steps 1, 2, and 3.

3/ The electrical measurements for table V of MIL-S-19500 are as follows:

a. Subgroup 2, see table III herein, steps 1, 2, 3, and 4.

b. Subgroup 3, (DO-13 devices only), see table III herein, steps 1, 2, and 3.

b. Subgroup 6, see table III herein, steps 1, 2, and 3.



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TABLE IV. Test ratings for diodes, types 1N3821A through 1N3828A and 1N3016B through 1N3051B (5 percent tolerance).

| Col 1 | Col 2 | Col 3 | Col 4 | Col 5 | Col 6 | Col 7 | Col 8 | Col 9 | Col 10 | Col 11 | Col 12 | | Col 13 | | Col 14 | Col 15 | Col 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Voltage group 1/ 2/ | $V_Z$ Nom 2/ | $V_Z$ Min 2/ | $V_Z$ Max 2/ | $I_Z$ Test current | $Z_Z$ Imped-ance | $Z_{ZK}$ Knee Imped-ance | $I_{ZM}$ Max dc current | $V_Z$ (reg) | $I_R$ (Surge) $T_A$ = 25°C | $V_R$ Reverse voltage | $I_R$ Reverse current dc μA | μA | $I_{R1}$ Reverse current dc μA | μA | (-1 only) $I_{R2}$ Reverse current dc; $T_A$ =+150°C | $\alpha V_Z$ Temperature coefficient | $I_{ZK}$ Test current |
| | Volts | Volts | Volts | mA | ohms | ohms | mA | Volts | mA | Volts | Non -1 | -1 | Non -1 | -1 | μA | %/°C | mA |
| 1N3821A | 3.3 | 3.14 | 3.46 | 76 | 10 | 400 | 276 | 1.00 | 1380 | 1 | 100 | 100 | 200 | 150 | 200 | -0.075 | 1.0 |
| 1N3822A | 3.6 | 3.42 | 3.78 | 69 | 10 | 400 | 252 | 0.80 | 1260 | 1 | 100 | 75 | 200 | 100 | 150 | - .070 | 1.0 |
| 1N3823A | 3.9 | 3.71 | 4.09 | 64 | 9 | 400 | 238 | 0.75 | 1190 | 1 | 50 | 25 | 100 | 40 | 100 | - .060 | 1.0 |
| 1N3824A | 4.3 | 4.09 | 4.51 | 58 | 9 | 400 | 213 | 0.70 | 1070 | 1 | 10 | 5 | 20 | 10 | 50 | - .050 | 1.0 |
| 1N3825A | 4.7 | 4.47 | 4.93 | 53 | 8 | 500 | 194 | 0.60 | 970 | 1 | 10 | 5 | 20 | 10 | 50 | ± .025 | 1.0 |
| 1N3826A | 5.1 | 4.85 | 5.35 | 49 | 7 | 550 | 178 | 0.50 | 890 | 1 | 10 | 3 | 20 | 6 | 50 | ± .030 | 1.0 |
| 1N3827A | 5.6 | 5.32 | 5.88 | 45 | 5 | 600 | 162 | 0.40 | 810 | 2 | 10 | 3 | 20 | 6 | 50 | + .040 | 1.0 |
| 1N3828A | 6.2 | 5.89 | 6.51 | 41 | 2 | 700 | 146 | 0.30 | 730 | 3 | 10 | 3 | 20 | 6 | 50 | + .050 | 1.0 |
| 1N3016B | 6.8 | 6.46 | 7.14 | 37 | 3.5 | 700 | 140 | 0.30 | 740 | 5.2 | 150 | 5.0 | 300 | 10 | 50 | + .057 | 1.0 |
| 1N3017B | 7.5 | 7.13 | 7.87 | 34 | 4.0 | 700 | 125 | 0.35 | 680 | 5.7 | 100 | 5.0 | 200 | 10 | 50 | + .061 | .5 |
| 1N3018B | 8.2 | 7.79 | 8.61 | 31 | 4.5 | 700 | 115 | 0.40 | 600 | 6.2 | 50 | 5.0 | 100 | 10 | 50 | + .065 | .5 |
| 1N3019B | 9.1 | 8.65 | 9.55 | 28 | 6.0 | 700 | 105 | 0.45 | 540 | 6.9 | 25 | 5.0 | 50 | 10 | 50 | + .068 | .5 |
| 1N3020B | 10 | 9.5 | 10.5 | 25 | 7 | 700 | 95 | 0.50 | 480 | 7.6 | 25 | 5.0 | 50 | 10 | 50 | + .071 | .25 |
| 1N3021B | 11 | 10.45 | 11.55 | 23 | 8 | 700 | 85 | 0.55 | 420 | 8.4 | 10 | 1.0 | 20 | 4 | 10 | + .073 | .25 |
| 1N3022B | 12 | 11.40 | 12.60 | 21 | 9 | 700 | 80 | 0.60 | 400 | 9.1 | 10 | 1.0 | 20 | 4 | 10 | + .076 | .25 |
| 1N3023B | 13 | 12.35 | 13.65 | 19 | 10 | 700 | 74 | 0.65 | 370 | 9.9 | 10 | .5 | 20 | 2 | 10 | + .079 | .25 |
| 1N3024B | 15 | 14.25 | 15.75 | 17 | 14 | 700 | 63 | 0.75 | 320 | 11.4 | 10 | .5 | 20 | 2 | 10 | + .082 | .25 |
| 1N3025B | 16 | 15.20 | 16.80 | 15.5 | 16 | 700 | 60 | 0.80 | 300 | 12.2 | 10 | .5 | 20 | 2 | 10 | + .083 | .25 |
| 1N3026B | 18 | 17.10 | 18.90 | 14.0 | 20 | 750 | 52 | 0.83 | 260 | 13.7 | 10 | .5 | 20 | 2 | 10 | + .085 | .25 |
| 1N3027B | 20 | 19.0 | 21.0 | 12.5 | 22 | 750 | 47 | 0.95 | 240 | 15.2 | 10 | .5 | 20 | 2 | 10 | + .086 | .25 |
| 1N3028B | 22 | 20.9 | 23.1 | 11.5 | 23 | 750 | 43 | 1.0 | 210 | 16.7 | 10 | .5 | 20 | 2 | 10 | + .087 | .25 |
| 1N3029B | 24 | 22.8 | 25.2 | 10.5 | 25 | 750 | 40 | 1.1 | 200 | 18.2 | 10 | .5 | 20 | 2 | 10 | + .088 | .25 |
| 1N3030B | 27 | 25.7 | 28.3 | 9.5 | 35 | 750 | 34 | 1.3 | 170 | 20.6 | 10 | .5 | 20 | 2 | 10 | + .090 | .25 |
| 1N3031B | 30 | 28.5 | 31.5 | 8.5 | 40 | 1000 | 31 | 1.4 | 160 | 22.8 | 10 | .5 | 20 | 2 | 10 | + .092 | .25 |
| 1N3032B | 33 | 31.4 | 34.6 | 7.5 | 45 | 1000 | 28 | 1.5 | 150 | 25.1 | 10 | .5 | 20 | 2 | 10 | + .091 | .25 |
| 1N3033B | 36 | 34.2 | 37.8 | 7.0 | 50 | 1000 | 26 | 1.7 | 130 | 27.4 | 10 | .5 | 20 | 2 | 10 | + .093 | .25 |
| 1N3034B | 39 | 37.1 | 40.9 | 6.5 | 60 | 1000 | 23 | 1.8 | 110 | 29.7 | 10 | .5 | 20 | 2 | 10 | + .094 | .25 |
| 1N3035B | 43 | 40.9 | 45.1 | 6.0 | 70 | 1500 | 21 | 1.9 | 100 | 32.7 | 10 | .5 | 20 | 2 | 10 | + .095 | .25 |
| 1N3036B | 47 | 44.7 | 49.3 | 5.5 | 80 | 1500 | 19 | 2.1 | 95 | 35.8 | 10 | .5 | 20 | 2 | 10 | + .095 | .25 |
| 1N3037B | 51 | 48.5 | 53.5 | 5.0 | 95 | 1500 | 18 | 2.3 | 90 | 38.8 | 10 | .5 | 20 | 2 | 10 | + .096 | .25 |
| 1N3038B | 56 | 53.2 | 58.8 | 4.5 | 110 | 2000 | 17 | 2.5 | 85 | 32.6 | 10 | .5 | 20 | 2 | 10 | + .096 | .25 |

See footnotes at end of table.





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TABLE IV. Test ratings for diodes, types 1N3821A through 1N3828A and 1N3016B through 1N3051B (5 percent tolerance).- Continued

| Col 1 | Col 2 | Col 3 | Col 4 | Col 5 | Col 6 | Col 7 | Col 8 | Col 9 | Col 10 | Col 11 | Col 12 | | Col 13 | | Col 14 | Col 15 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Voltage group 1/ 2/ | $V_Z$ Nom 2/ | $V_Z$ Min 2/ | $V_Z$ Max 2/ | $I_Z$ Test current | $Z_Z$ Impedance | $Z_{ZK}$ (Knee Impedance) | $I_{ZM}$ Max dc current | $V_Z$ (reg) | $I_{ZSM}$ (Surge) $T_A$ = 25°C | $V_R$ Reverse voltage | $I_R$ Reverse current dc μA | μA | $I_{R1}$ Reverse current dc μA | μA | (-1 only) $I_{R2}$ Reverse current dc; $T_A$ =+150°C | $\alpha V_Z$ Temperature Coefficient | $I_{ZK}$ Test current |
| | Volts | Volts | Volts | mA | ohms | ohms | mA | Volts | mA | Volts | Non -1 | -1 | Non -1 | -1 | μA | %/°C | mA |
| 1N3039B | 63 | 58.95 | 65.1 | 4.0 | 125 | 2000 | 15 | 2.7 | 75 | 47.1 | 10 | .5 | 20 | 2 | 10 | + 0.097 | .25 |
| 1N3040B | 68 | 64.60 | 71.4 | 3.7 | 150 | 2000 | 14 | 3.0 | 70 | 51.7 | 10 | .5 | 20 | 2 | 10 | + .097 | .25 |
| 1N3041B | 75 | 71.35 | 78.7 | 3.3 | 175 | 2000 | 12 | 3.3 | 63 | 56.0 | 10 | .5 | 20 | 2 | 10 | + .098 | .25 |
| 1N3042B | 82 | 77.95 | 86.1 | 3.0 | 200 | 3000 | 11 | 3.6 | 58 | 62.2 | 10 | .5 | 20 | 2 | 10 | + .098 | .25 |
| 1N3043B | 91 | 86.5 | 95.5 | 2.8 | 250 | 3000 | 10 | 4.0 | 50 | 69.2 | 10 | .5 | 20 | 2 | 10 | + .099 | .25 |
| 1N3044B | 100 | 95.0 | 105.0 | 2.5 | 350 | 3000 | 9 | 4.4 | 45 | 76.0 | 10 | .5 | 20 | 2 | 10 | + .100 | .25 |
| 1N3045B | 110 | 104.5 | 115.5 | 2.3 | 450 | 4000 | 8.3 | 5.0 | 42 | 83.6 | 10 | .5 | 20 | 2 | 10 | + .100 | .25 |
| 1N3046B | 120 | 114 | 126.0 | 2.0 | 550 | 4500 | 8.0 | 5.5 | 40 | 91.2 | 10 | .5 | 20 | 2 | 10 | + .100 | .25 |
| 1N3047B | 130 | 123.5 | 136.5 | 1.9 | 700 | 5000 | 6.9 | 6.0 | 35 | 98.8 | 10 | .5 | 20 | 2 | 10 | + .100 | .25 |
| 1N3048B | 150 | 142.5 | 157.5 | 1.7 | 1000 | 6000 | 5.7 | 7.0 | 29 | 114.0 | 10 | .5 | 20 | 2 | 10 | + .100 | .25 |
| 1N3049B | 160 | 152 | 168.0 | 1.6 | 1100 | 6500 | 5.4 | 8.0 | 27 | 121.6 | 10 | .5 | 20 | 2 | 10 | + .100 | .25 |
| 1N3050B | 180 | 171 | 189.0 | 1.4 | 1200 | 7000 | 4.9 | 10.0 | 25 | 136.8 | 10 | .5 | 20 | 2 | 10 | + .100 | .25 |
| 1N3051B | 200 | 190 | 210.0 | 1.2 | 1500 | 8000 | 4.6 | 12.0 | 23 | 152.0 | 10 | .5 | 20 | 2 | 10 | + .100 | .25 |

1/ Ratings also apply to dash one and surface mount devices unless otherwise noted.

2/ 5 percent voltage tolerances are shown in table.
For 2 percent tolerance ("C" suffix; for "-1" suffix and JANHC only), column 3 is 2 percent less than column 2, column 4 is 2 percent more than column 2.
For 1 percent tolerance ("D" suffix; for "-1" suffix and JANHC only), column 3 is 1 percent less than column 2, column 4 is 1 percent more than column 2.





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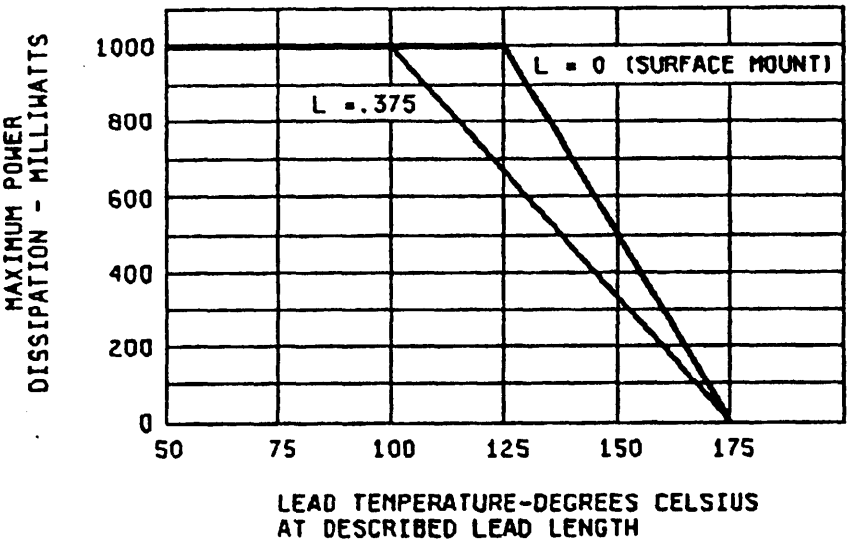



FIGURE 7. Maximum power versus lead temperature and lead length (see 6.4).



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5. PACKAGING

5.1 Packaging requirements. Packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.
b. Lead finish (see 3.3.1).
c. For die acquisition, the JANHC letter version should be specified (see figures 4 and 5).
d. Product assurance level and type designation.

6.3 Substitution information. Devices covered by this specification are substitutable for the manufacturer's and user's part number. This information in no way implies that manufacturers' part numbers are suitable as a substitute for the military part number.

6.3.1 Substitutability of 2 percent and 1 percent tolerance devices. Devices of tighter tolerance are a direct one way substitute for the looser tolerance devices (example: JANTX1N3821D-1 substitutes for JANTX1N3821A-1).

6.4 Maximum power versus lead temperature. Typical maximum power rating as a function of lead temperature for various lead lengths is shown on figure 7.

6.5 Suppliers of JANHC die. The qualified JANHC suppliers with the applicable letter version (example JANHCA1N3821A) will be identified on the QPL.

| JANHC ordering information | | |
|---|---|---|
| PIN | Manufacturer CAGE | |
| | 55801 | 12954 |
| 1N3821A through 1N3828A | JANHCA1N3821A through JANHCA1N3828A | JANHCB1N3821A through JANHCB1N3828A |
| 1N3016B through 1N3051B | JANHCA1N3016B through JANHCA1N3051B | JANHCB1N3016B through JANHCB1N3051B |

NOTE: C and D tolerance suffix are applicable to JANHC.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.



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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, AV, MI, SM
Navy - AS, CG, MC
Air Force - 13, 19, 85, 99
DLA - ES

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5961-1538)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

**INSTRUCTIONS**

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER<br>MIL-S-19500/115H | 2. DOCUMENT DATE (YYMMDD)<br>20 December 1993 |
|---|---|---|

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, DIODE, SILICON, VOLTAGE REGULATOR TYPES 1N3821A THROUGH 1N3828A, 1N3016B THROUGH 1N3051B, 1N3821A-1 THROUGH 1N3828A-1, 1N3016B-1 THROUGH 1N3051B-1, 1N3821AUR-1 THROUGH 1N3828AUR-1, 1N3016BUR-1 THROUGH 1N3051BUR-1, PLUS C- AND D- TOLERANCE SUFFIX, JAN, JANTX, JANTXV, JANHC, AND JANKC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (Last, First, Middle initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON<br>(If applicable) | 7. DATE SUBMITTED<br>(YYMMDD) |

8. PREPARING ACTIVITY

| a. NAME<br>Alan Barone | b. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(513)-296-6048 | (2) AUTOVON<br>986-6048 |
|---|---|---|
| c. ADDRESS (Include Zip Code)<br>DESC-ECT<br>Dayton, OH 45444-5270 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 | |



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